Addressing the global misinformation crisis one flag at a time

RedFlag

The future of fact checking

By Team Bronze
Old Dominion University
CS 410
2/14/2026

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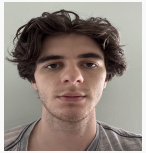
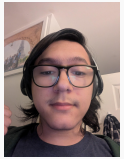
Our *team*



Ayden Garcia

TITLE: Team Lead

agarc046@odu.edu



James Moore

TITLE: Team member

jmoor072@odu.edu

Cameron Zell

TITLE: Team member

czell002@odu.edu

Trey Leverett

TITLE: Team member

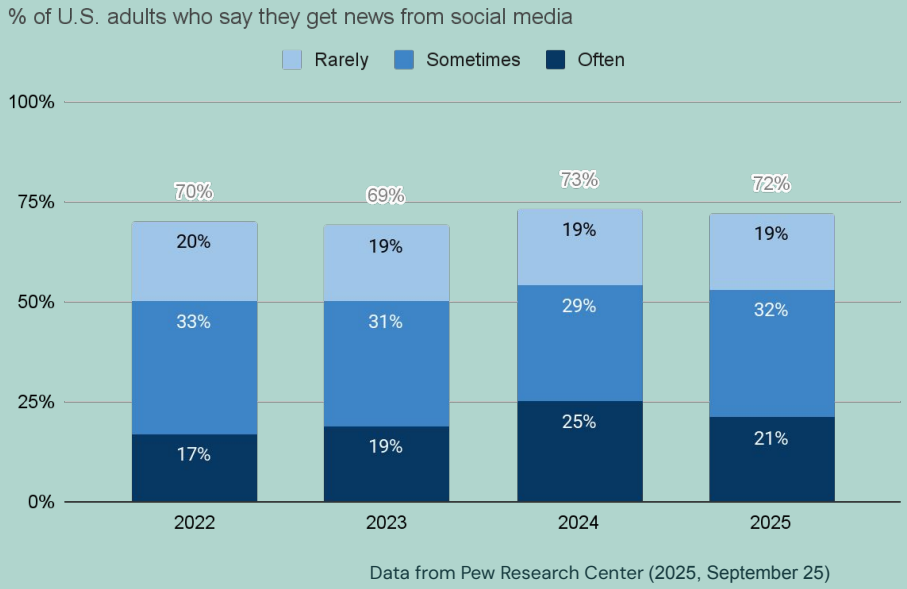
tleve002@odu.edu

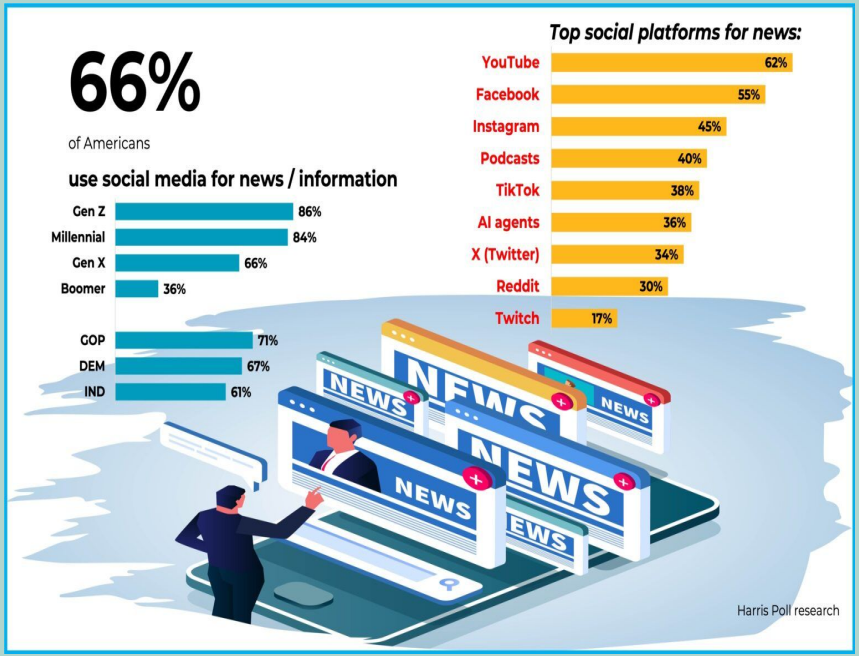
Riley Reynolds

TITLE: Team Member

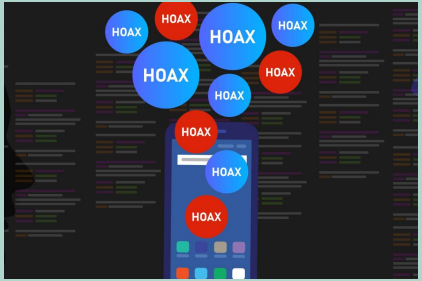
areyn017@odu.edu

Social Media as News



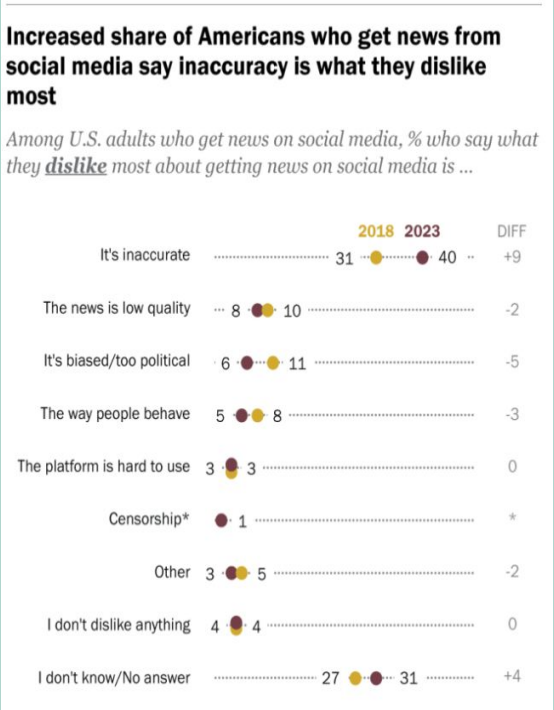


Misinformation in our Digital Ecosystem



- 94% of people report encountering misinformation online (The Alan Turing Institute, 2024)
- 38.2% of U.S. news consumers report unknowingly sharing fake news. (Redline Digital, 2024)

Distrust is growing



Pew Research Center. (2024, February 7)

- Concerns about inaccurate information on social media are rising (Pew Research Center, 2024)
- 40% of Americans who get news from social media cite inaccuracy as their biggest concern. (Pew Research Center, 2024)
- Only 37% say they trust information from social media platforms. (Pew Research Center, 2023)

(Why misinfo spreads)

(Why aren’t current options working)

Why does this matter?

When misinformation spreads:

- Public trust declines (Pew Research Center, 2023)
- Users feel concerned but lack effective tools (Pew Research Center, 2024)
- False information spreads unintentionally (Redline Digital, 2024)

RedFlag

Problem Characteristics (THE GAP)

Convenience

Fact-checking often requires manual searching outside the platform.

Awareness

Users often receive no real-time credibility signals while browsing.

Motivation

Platforms and algorithms encourage instant sharing without friction.

Fragmentation

Many existing verification tools are platform-specific and not unified.

The Solution

RedFlag is a cross-platform browser extension that automatically detects and flags potentially misleading content in real time, providing credibility indicators and source links directly within the user's browsing experience.

What it Will Do

How will RedFlag work?

Integrated Detection

- Integrated by a chrome extension to scan content automatically

Real-Time Detection

- Highlights questionable information immediately

Frictionless & Complete

- Highlights require no user input
- Can deliver detailed overview on request

RedFlag

Closing THE GAP

Convenience	Awareness	Motivation	Fragmentation
Real Time Integrated Detection	Real-Time detection	Frictionless Detection System	Multi-Platform capabilities

What it Will Not Do

What will RedFlag's Limitations be?

Unbiased political alignment

- RedFlag will not support any specific political standpoint
- No ideological labeling
- Evidence based evaluation

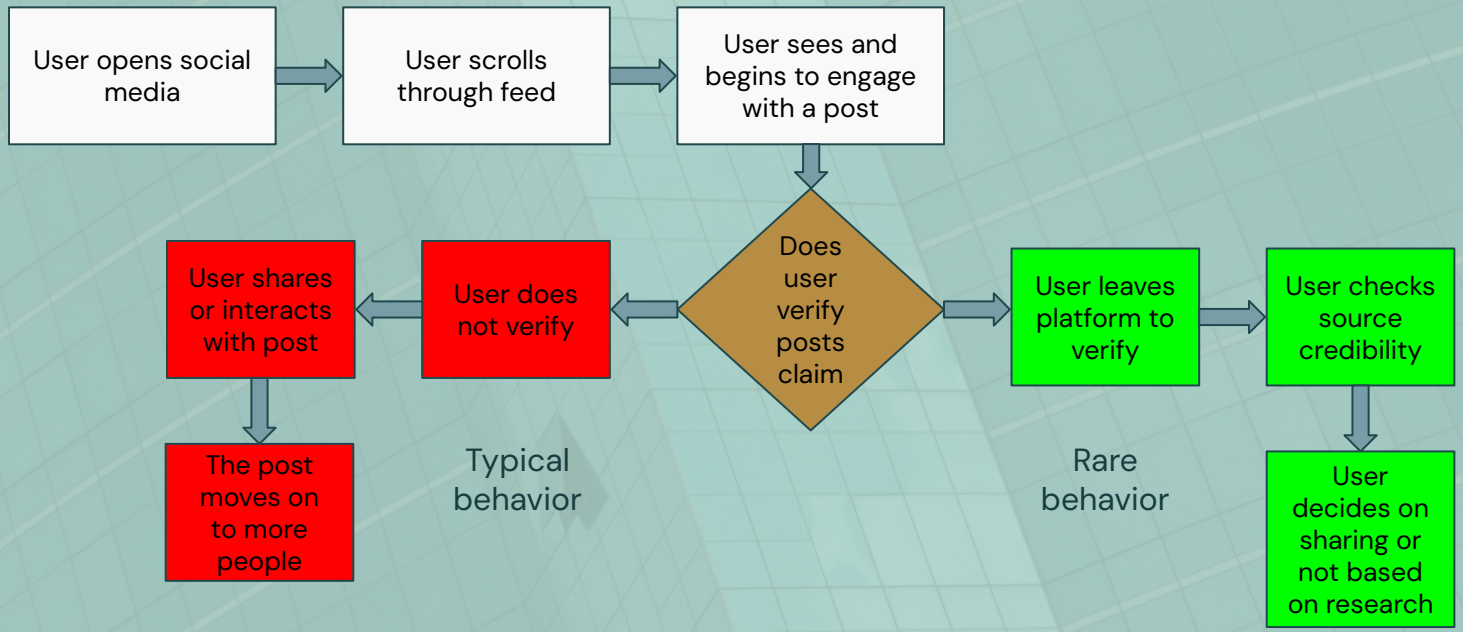
Uninterrupted flow

- Pop-ups would interrupt user's browsing experience
- Highlighting text is significantly less intrusive

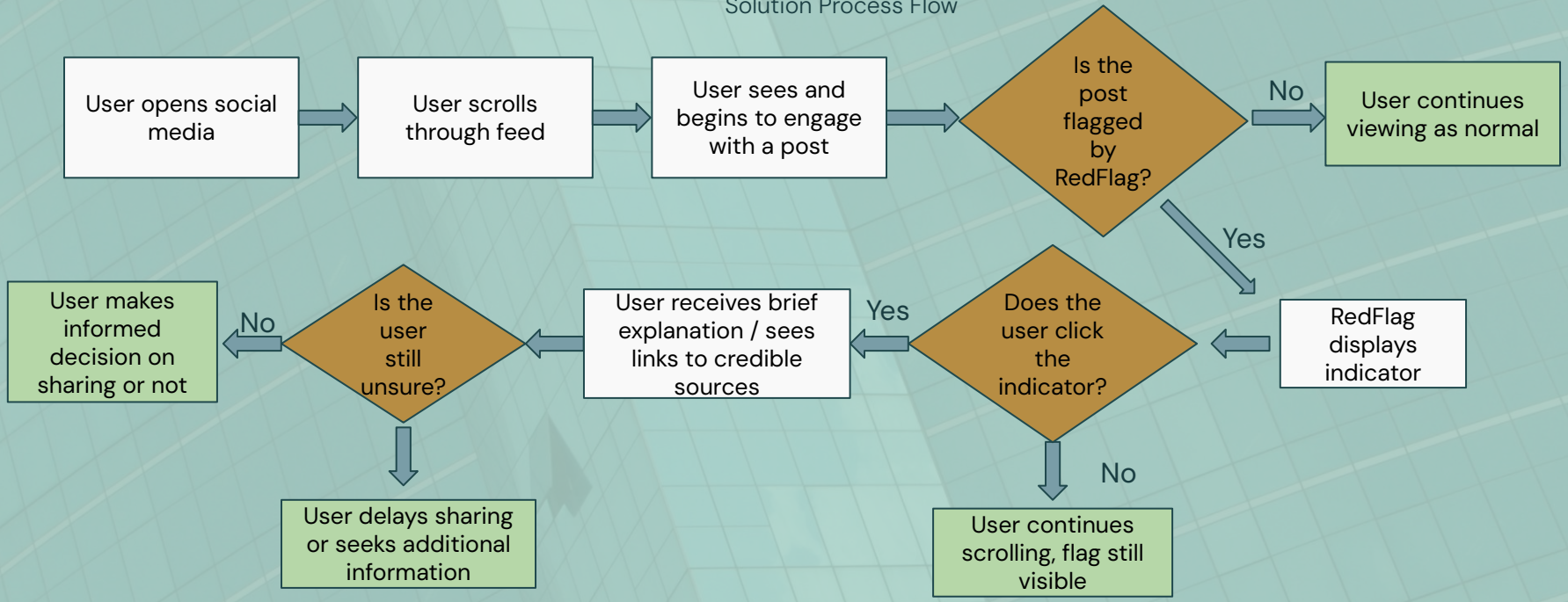
Does NOT replace professional fact-checkers

- Supports decision-making without dictating it
- Optional context is provided upon the user's request
- Supports journalism, but does not substitute it

Problem Process Flow



Solution Process Flow



COMPETITIVE LANDSCAPE

Competitors

	Google Fact Check Explorer	News Guard	Community Notes	AI
FEATURES	• Search for stories by keyword, person, or image that have been debunked • Fact check other stories yourself	• Rates trustworthiness of a news source on a scale from 1-100 • Team of journalists fact checking	• Allows users to add contextual explanations or fact checks to posts • Other contributors rate notes for helpfulness, only helpful notes are shown	• Ask AI the veracity of a claim/source (e.g. @Grok is this true?) • AI uses its dataset to respond
STRENGTHS	• Quickly able to search for rumors or claims being made online • Aggregate source of online disinformation	• Real human input on trustworthiness • Partnered with organizations for better fact checking	• Posts remain visible with extra context added • Uses a diverse group of users to reduce bias	• Ease of use- users already familiar with service, easily available, immediate response • Low manpower required
WEAKNESSES	• Allows biased sources to fact check other sources • Does not provide input on every claim made online • No verified source of claims	• Possible bias due to human fact checkers •	• Slow response time makes notes appear long after a misleading post has spread • Organized groups can attempt to manipulate note ratings	• Frequently incorrect • Confidently wrong, fooling users • Dataset is fallible

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Development Tools

Languages

Test Frameworks

Version Control

Documentation

Major Functional Components

Browser Extension UI

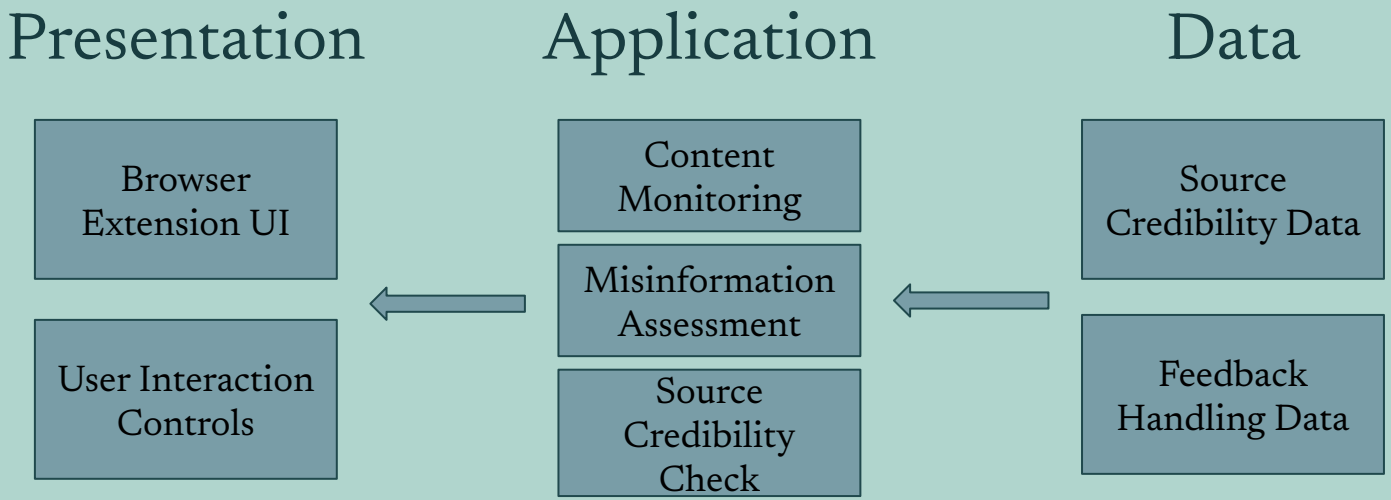
User Interaction Controls

Content Monitoring

Misinformation Assessment

Source Credibility Data

Feedback Handling Data



Major Functional Components Diagram

Glossary

- Friction - something that causes an application to not be seamless while working.

Risks

Appendix

Real World Product vs Prototype Table

Elevator Pitch

RedFlag is a browser extension designed to reduce the spread of online misinformation. It runs in the background while users browse social media and news sites and automatically highlights potentially false or misleading content directly on the screen. Instead of relying on users to fact-check information themselves, RedFlag brings awareness to misinformation at the moment it's encountered. With one click, users can view a short explanation and links to credible sources, helping them make informed decisions before believing or sharing content.

The problem plaguing online news

Despite the vastly popular concern in regarding digital inaccuracy, social media users frequently propagate misinformation because existing verification methods are inconvenient and require active user intervention. There is a critical need for a passive, real-time tool that bridges the gap between content consumption and fact-checking for daily social media users.

Who does this affect?

- Users of social media
  - Teens
    - 85% of young people now get their news primarily from social media. (Stagwell, 2025)
  - 55+
    - On Twitter, 80% of fake news was shared by users over age 50 (Marshall, 2025)